

IMPROVEMENT IN FOOD RESOURCES

Class 9 | Science | Chapter 15

NEED FOR IMPROVEMENT IN FOOD PRODUCTION

With a rising population, food production must increase in a sustainable way -- without harming the environment.

Two broad approaches: crop production improvement, and animal husbandry/livestock production improvement.

CROP IMPROVEMENT -- 3 MAIN APPROACHES

Approach	Meaning
Crop variety improvement	Breeding new, better varieties (hybridisation, genetic engineering)
Crop production improvement	Better management practices -- irrigation, cropping patterns, nutrients
Crop protection management	Protecting crops from weeds, pests, and diseases

DESIRABLE TRAITS IN IMPROVED VARIETIES

- Higher yield
- Improved quality (e.g. protein content)
- Biotic & abiotic resistance (disease, drought, salinity)
- Shorter maturity duration
- Wider adaptability

CROP PRODUCTION MANAGEMENT

Nutrients, Irrigation & Cropping Patterns

NUTRIENT MANAGEMENT

Source	Examples
Manure	Organic matter from plant/animal waste -- improves soil structure
Fertilisers	Commercially produced nutrients (N, P, K) -- give quick, high yield but excess harms soil & water

Manure is bulky, has less nutrient content but improves soil texture. Fertilisers are nutrient-rich but do not add organic matter.

IRRIGATION

Water supply for crops from sources like wells, canals, rivers, tanks. Modern methods (sprinkler, drip irrigation) use water more efficiently.

CROPPING PATTERNS

Pattern	Meaning
Mixed cropping	Growing 2+ crops together to reduce risk of total crop failure
Intercropping	Growing 2+ crops in a defined row pattern for higher productivity
Crop rotation	Growing different crops in succession on the same field to maintain soil fertility

CROP PROTECTION & ANIMAL HUSBANDRY

Weeds, Pests & Livestock

CROP PROTECTION MANAGEMENT

Weeds

Unwanted plants growing along with the crop, competing for nutrients/water/light -- controlled by mechanical removal or weedicides.

Pests & Diseases

Insects, rodents, fungi/bacteria/viruses attack crops at various stages -- controlled using resistant varieties & (limited) pesticides, and good agricultural practices.

ANIMAL HUSBANDRY

Scientific management of livestock (cattle, poultry, fish etc.) for food and other products.

Cattle Farming

Milch (dairy) animals for milk; Draught animals for farm labour. Managed via proper shelter, feed & disease control.

Poultry Farming

Rearing of domestic fowl for eggs (layers) and meat (broilers).

Fish Production

Includes marine (capture) fisheries and inland (culture) fisheries such as aquaculture.

BEE-KEEPING & SUMMARY

Apiculture + Quick Revision

BEE-KEEPING (APICULTURE)

Rearing honeybees for honey and beeswax -- also helps pollinate crops, improving yield.

A good honeybee variety should: produce more honey, sting less, stay in a given hive for a long duration.

CHAPTER SUMMARY -- KEY TERMS

- **Crop variety improvement** -> breeding better yielding, resistant varieties
- **Manure vs Fertiliser** -> organic bulk vs concentrated nutrients
- **Mixed cropping/Intercropping/Crop rotation** -> cropping pattern strategies
- **Animal husbandry** -> scientific management of livestock
- **Apiculture** -> bee-keeping for honey & pollination

ONE-LINE RECAP FOR REVISION

Food security needs both better crop varieties AND better management.

Balance fertilisers with manure to protect long-term soil health.

Animal husbandry & apiculture add to food resources beyond crops.

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